Describing Quantum States

A Complete Guide to State Description in Quantum Mechanics

Central Question

How do we describe the state of the degrees of freedom of a system in quantum mechanics?

1 Introduction: The Nature of Quantum States

In classical mechanics, we describe a particle's state by specifying its position $\vec{r}(t)$ and momentum $\vec{p}(t)$ at time t. These determine the trajectory completely.

Key Idea

In quantum mechanics, the state description is fundamentally probabilistic. Instead of definite values, we work with probability amplitudes that tell us the likelihood of finding specific measurement outcomes.

The quantum state contains **all possible information** about the system, but this doesn't mean we can predict every measurement with certainty—only the probabilities of different outcomes.

2 The Fundamental Framework

2.1 Two Levels of Knowledge

Quantum mechanics recognizes two fundamentally different types of state descriptions:

Pure States
Complete quantum knowledge
State vector $|\psi\rangle$

↓ Adding classical uncertainty

Mixed States
Statistical mixture
Density matrix $\hat{\rho}$

3 Pure States: Complete Quantum Information

Definition 3.1 (Pure State)**.** A **pure state** represents the maximum knowledge we can have about a quantum system. It is described by a normalized state vector $|\psi\rangle$ living in a Hilbert space $\mathcal{H}$:

$$|\psi\rangle \in \mathcal{H}, \quad \langle\psi|\psi\rangle = 1 \tag{1}$$

3.1 What Does the State Vector Tell Us?

The state vector $|\psi\rangle$ encodes probability amplitudes. For any measurement outcome corresponding to state $|\phi\rangle$:

$$\text{Probability amplitude: } \langle\phi|\psi\rangle \quad \Rightarrow \quad \text{Probability: } P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 \tag{2}$$

Important

The Born Rule: The probability of measuring the system in state $|\phi\rangle$ when it's prepared in state $|\psi\rangle$ is given by the squared magnitude of their inner product.

3.2 Key Properties of Pure States

1. **Normalization:** The total probability must be 1: $\langle\psi|\psi\rangle = 1$
2. **Superposition:** If $|\psi_1\rangle$ and $|\psi_2\rangle$ are valid states, so is:

$$|\psi\rangle = \alpha|\psi_1\rangle + \beta|\psi_2\rangle, \quad |\alpha|^2 + |\beta|^2 = 1 \tag{3}$$

3. **Phase Invariance:** States $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are physically identical (global phase has no observable effect)
4. **Linearity:** The state space is a complex vector space

3.3 Representations in Different Bases

The same state $|\psi\rangle$ can be expressed in different bases, revealing different information:

Discrete Basis (e.g., Energy Eigenstates)

$$|\psi\rangle = \sum_n c_n |E_n\rangle, \quad c_n = \langle E_n|\psi\rangle, \quad \sum_n |c_n|^2 = 1 \tag{4}$$

Here $|c_n|^2$ gives the probability of measuring energy E_n.

Position Representation (Wavefunction)

$$\psi(x) = \langle x|\psi\rangle \tag{5}$$

The wavefunction $\psi(x)$ is what we often call "the quantum state," but it's really just one representation of the abstract state vector $|\psi\rangle$.

Key Idea

$|\psi(x)|^2$ gives the probability density of finding the particle at position x. The total probability is:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1 \tag{6}$$

Momentum Representation

$$\tilde{\psi}(p) = \langle p|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-ipx/\hbar} \psi(x) \, dx \tag{7}$$

This is the Fourier transform of the position wavefunction. $|\tilde{\psi}(p)|^2$ gives the probability density in momentum space.

3.4 Example: The Spin-1/2 System

Example 3.1 (Spin-1/2 Particle)**.** Consider an electron's spin (intrinsic angular momentum). The Hilbert space is 2-dimensional: $\mathcal{H} = \mathbb{C}^2$.

Basis states: Spin up and down along the z-axis:

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{8}$$

General pure state:

$$|\psi\rangle = \alpha \, |\uparrow\rangle + \beta \, |\downarrow\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1 \tag{9}$$

Physical interpretation:

- $|\alpha|^2$ = probability of measuring spin-up along z
- $|\beta|^2$ = probability of measuring spin-down along z

Bloch sphere representation: Every pure spin-1/2 state can be written as:

$$|\psi(\theta, \phi)\rangle = \cos\frac{\theta}{2} \, |\uparrow\rangle + e^{i\phi} \sin\frac{\theta}{2} \, |\downarrow\rangle \tag{10}$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ parameterize points on a sphere.

3.5 Multiple Degrees of Freedom

Real systems have multiple degrees of freedom. The total Hilbert space is a tensor product:

Particle with Spin in 3D Space

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{spatial}} \otimes \mathcal{H}_{\text{spin}} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2 \tag{11}$$

State representation:

$$|\Psi\rangle = \int d^3r \, [\psi_\uparrow(\vec{r}) \, |\vec{r}, \uparrow\rangle + \psi_\downarrow(\vec{r}) \, |\vec{r}, \downarrow\rangle] \tag{12}$$

Or as a spinor wavefunction:

$$\Psi(\vec{r}) = \begin{pmatrix} \psi_\uparrow(\vec{r}) \\ \psi_\downarrow(\vec{r}) \end{pmatrix} \tag{13}$$

Many-Particle Systems

N distinguishable particles:

$$|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_N \tag{14}$$

Identical particles require symmetry conditions:

- **Bosons** (integer spin): Symmetric states

$$|\Psi(\vec{r}_1, \vec{r}_2)\rangle = |\Psi(\vec{r}_2, \vec{r}_1)\rangle \tag{15}$$

- **Fermions** (half-integer spin): Antisymmetric states (Pauli exclusion principle)

$$|\Psi(\vec{r}_1, \vec{r}_2)\rangle = - |\Psi(\vec{r}_2, \vec{r}_1)\rangle \tag{16}$$

4 Mixed States: Statistical Ensembles

Pure states assume we have complete quantum information. But what if we don't?

Key Idea

When we have statistical uncertainty about which pure state the system is in, we need the density matrix formalism. This describes **mixed states**.

Definition 4.1 (Mixed State)**.** A **mixed state** represents incomplete knowledge. It describes a statistical ensemble where the system is in pure state $|\psi_i\rangle$ with classical probability p_i:

$$\hat{\rho} = \sum_i p_i \, |\psi_i\rangle \langle\psi_i| \, , \quad p_i \geq 0, \quad \sum_i p_i = 1 \tag{17}$$

4.1 The Density Matrix Framework

The density operator $\hat{\rho}$ provides a unified description that encompasses both pure and mixed states.

Fundamental Properties

$$\boxed{\begin{array}{l}\text{1. Hermitian:}\quad \hat{\rho}^\dagger = \hat{\rho} \\ \text{2. Normalized:}\quad \mathrm{Tr}(\hat{\rho}) = 1 \\ \text{3. Positive:}\quad \hat{\rho} \geq 0 \text{ (all eigenvalues} \geq 0)\end{array}} \tag{18}$$

Pure vs. Mixed: The Purity Test

Important

Purity criterion:

$$\mathcal{P} = \mathrm{Tr}(\hat{\rho}^2) = \begin{cases} 1 & \text{pure state} \\ < 1 & \text{mixed state} \end{cases} \tag{19}$$

For a pure state $|\psi\rangle$: $\hat{\rho} = |\psi\rangle\langle\psi|$, so $\hat{\rho}^2 = \hat{\rho}$ (idempotent).

4.2 Example: Unpolarized vs. Polarized Spin

Example 4.1 (Comparing Pure and Mixed Spin States)**. Pure state:** Spin pointing in the +x direction:

$$|+x\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) \tag{20}$$

Density matrix:

$$\hat{\rho}_{\text{pure}} = |+x\rangle\langle+x| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \tag{21}$$

Purity: $\mathrm{Tr}(\hat{\rho}^2_{\text{pure}}) = 1$

Mixed state: Unpolarized spin (50% up, 50% down):

$$\hat{\rho}_{\text{mixed}} = \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{22}$$

Purity: $\mathrm{Tr}(\hat{\rho}^2_{\text{mixed}}) = \frac{1}{2} < 1$

Key difference: Both give 50% probability for measuring up or down along z. But:

- Pure state: Measuring along x gives definite result (+x with 100% probability)
- Mixed state: Measuring along x gives 50-50 result (no preferred direction)

The off-diagonal terms (coherences) distinguish them!

4.3 Sources of Mixed States

Mixed states arise in several physical situations:

1. Classical Ignorance

We know the system is in one of several pure states, but we don't know which:

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i| \tag{23}$$

2. Quantum Entanglement

When we trace out part of an entangled system, the remaining subsystem is generally mixed.

Example 4.2 (Entanglement-Induced Mixing)**.** Consider a Bell state (maximally entangled):

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{24}$$

This is a **pure state** of the composite system. But if we look only at subsystem A:

$$\hat{\rho}_A = \mathrm{Tr}_B(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbb{I} \tag{25}$$

Subsystem A is **maximally mixed**! This is the signature of entanglement.

3. Thermal Equilibrium

At finite temperature T, the system is in a thermal mixed state:

$$\hat{\rho}_{\text{thermal}} = \frac{1}{Z}e^{-\beta\hat{H}}, \quad Z = \mathrm{Tr}(e^{-\beta\hat{H}}), \quad \beta = \frac{1}{k_B T} \tag{26}$$

where Z is the partition function.

4. Decoherence

Interaction with an environment destroys quantum coherences, turning pure states into mixed states.

4.4 Spectral Decomposition

Any density matrix can be diagonalized:

$$\hat{\rho} = \sum_i \lambda_i |\phi_i\rangle\langle\phi_i| \tag{27}$$

where:

- $\lambda_i \geq 0$ are eigenvalues (occupation probabilities)
- $\sum_i \lambda_i = 1$ (normalization)
- $\{|\phi_i\rangle\}$ are orthonormal eigenstates

Maximally mixed state: Equal occupation of all states:

$$\hat{\rho}_{\text{max mixed}} = \frac{1}{d}\mathbb{I}, \quad d = \dim(\mathcal{H}) \tag{28}$$

Purity: $\mathrm{Tr}(\hat{\rho}^2) = \frac{1}{d}$ (minimum possible)

5 Observables and Measurements

5.1 Expectation Values

For any observable $\hat{A}$, the expectation value is:

$$\boxed{\langle \hat{A} \rangle = \begin{cases} \langle \psi | \hat{A} | \psi \rangle & \text{(pure state)} \\ \mathrm{Tr}(\hat{\rho}\hat{A}) & \text{(general)} \end{cases}} \tag{29}$$

The density matrix formula works for both pure and mixed states!

5.2 Projective Measurements

Consider measuring an observable with eigenstates $\{|a_i\rangle\}$ and eigenvalues $\{a_i\}$.

Measurement probabilities:

$$P(a_i) = \mathrm{Tr}(\hat{\rho}\hat{P}_i), \quad \hat{P}_i = |a_i\rangle \langle a_i| \tag{30}$$

Post-measurement state: If outcome a_i is obtained:

$$\hat{\rho}' = \frac{\hat{P}_i \hat{\rho} \hat{P}_i}{\mathrm{Tr}(\hat{\rho}\hat{P}_i)} \tag{31}$$

This is the **collapse** or **reduction** of the state.

6 Time Evolution

6.1 Pure State Evolution

Pure states evolve according to the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle \tag{32}$$

Formal solution:

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle, \quad \hat{U}(t) = e^{-i\hat{H}t/\hbar} \tag{33}$$

where $\hat{U}(t)$ is the unitary time evolution operator.

6.2 Density Matrix Evolution

The density matrix evolves via the von Neumann equation:

$$\boxed{\frac{\partial \hat{\rho}}{\partial t} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}]} \tag{34}$$

This is the quantum Liouville equation. Its solution is:

$$\hat{\rho}(t) = \hat{U}(t)\hat{\rho}(0)\hat{U}^\dagger(t) \tag{35}$$

Important

Unitary evolution preserves purity: if $\text{Tr}(\hat{\rho}^2(0)) = 1$, then $\text{Tr}(\hat{\rho}^2(t)) = 1$ for all t. To create mixed states from pure ones requires non-unitary processes (measurement, decoherence, tracing out subsystems).

7 Specific Degrees of Freedom

7.1 Position and Momentum

Position Space

The wavefunction:

$$\psi(x,t) = \langle x|\psi(t)\rangle \tag{36}$$

Physical meaning: $|\psi(x,t)|^2$ is the probability density for finding the particle at position x at time t.

Density matrix in position basis:

$$\rho(x,x',t) = \langle x|\hat{\rho}(t)|x'\rangle \tag{37}$$

- Diagonal: $\rho(x,x,t)$ = probability density
- Off-diagonal: $\rho(x,x',t)$ for $x \neq x'$ = quantum coherence between positions

Momentum Space

The momentum wavefunction:

$$\tilde{\psi}(p,t) = \langle p|\psi(t)\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-ipx/\hbar}\psi(x,t)\,dx \tag{38}$$

This is the Fourier transform of the position wavefunction.

7.2 Spin States

For a spin-s particle, the spin Hilbert space has dimension $2s+1$:

$$\mathcal{H}_{\text{spin}} = \mathbb{C}^{2s+1} \tag{39}$$

Spin-1/2: Two-dimensional, basis $\{|\uparrow\rangle, |\downarrow\rangle\}$
General state:

$$|\psi\rangle = \cos\frac{\theta}{2}|\uparrow\rangle + e^{i\phi}\sin\frac{\theta}{2}|\downarrow\rangle \tag{40}$$

Bloch vector representation: Any spin-1/2 density matrix can be written:

$$\hat{\rho} = \frac{1}{2}(\mathbb{I} + \vec{r}\cdot\vec{\sigma}) \tag{41}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are Pauli matrices and $|\vec{r}| \leq 1$.

- Pure states: $|\vec{r}| = 1$ (surface of Bloch sphere)
- Mixed states: $|\vec{r}| < 1$ (interior)
- Maximally mixed: $\vec{r} = 0$ (center)

7.3 Angular Momentum

States labeled by quantum numbers ℓ (total) and m (z-component):

$$|\ell, m\rangle, \quad m \in \{-\ell, -\ell+1, \ldots, \ell-1, \ell\} \tag{42}$$

Eigenvalue equations:

$$\hat{L}^2 |\ell, m\rangle = \hbar^2 \ell(\ell+1) |\ell, m\rangle \tag{43}$$

$$\hat{L}_z |\ell, m\rangle = \hbar m |\ell, m\rangle \tag{44}$$

General state:

$$|\psi\rangle = \sum_{m=-\ell}^{\ell} c_m |\ell, m\rangle, \quad \sum_m |c_m|^2 = 1 \tag{45}$$

7.4 Energy (Bound States)

For systems with discrete energy spectra:

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle \tag{46}$$

where $\hat{H} |E_n\rangle = E_n |E_n\rangle$.

Example 7.1 (Quantum Harmonic Oscillator)**.** Energy eigenvalues:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \ldots \tag{47}$$

General state:

$$|\psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle \tag{48}$$

Time evolution:

$$|\psi(t)\rangle = \sum_{n=0}^{\infty} c_n e^{-i\omega(n+1/2)t} |n\rangle \tag{49}$$

8 Summary: The Complete Picture

Type	**Description**	**Mathematical Form**	**Purity**
Pure State	Complete quantum knowledge	$\|\psi\rangle$ or $\hat{\rho} = \|\psi\rangle \langle\psi\|$	$\mathrm{Tr}(\hat{\rho}^2) = 1$
Mixed State	Statistical ensemble	$\hat{\rho} = \sum_i p_i \|\psi_i\rangle \langle\psi_i\|$	$\mathrm{Tr}(\hat{\rho}^2) < 1$
Maximally Mixed	Maximum ignorance	$\hat{\rho} = \frac{1}{d}\mathbb{I}$	$\mathrm{Tr}(\hat{\rho}^2) = \frac{1}{d}$

Key Idea

The Hierarchy of Quantum State Descriptions:

1. **State vector** $|\psi\rangle$: Best for pure states, specific representation
2. **Density matrix** $\hat{\rho}$: Universal framework for pure and mixed states
3. **Wavefunction** $\psi(x)$: Position representation of state vector

The density matrix formalism is the most general and is essential for:

- Open quantum systems (decoherence)
- Quantum statistical mechanics (thermal states)
- Quantum information (entanglement, partial information)
- Subsystems of composite systems

8.1 Key Takeaways

1. **Pure states** describe systems with complete quantum information; they're represented by state vectors $|\psi\rangle$ in Hilbert space.
2. **Mixed states** describe statistical ensembles where we have incomplete information; they require density matrices $\hat{\rho}$.
3. **The density matrix** is the universal description that works for both pure and mixed states.
4. **Purity** $\mathrm{Tr}(\hat{\rho}^2)$ distinguishes pure ($= 1$) from mixed (< 1) states.
5. **Different bases** (position, momentum, energy) give different representations of the same quantum state.
6. **Composite systems** have Hilbert spaces built from tensor products of individual spaces.
7. **Entanglement** can make subsystems mixed even when the total system is pure.
8. **Time evolution** is unitary and preserves purity; mixing requires non-unitary processes.